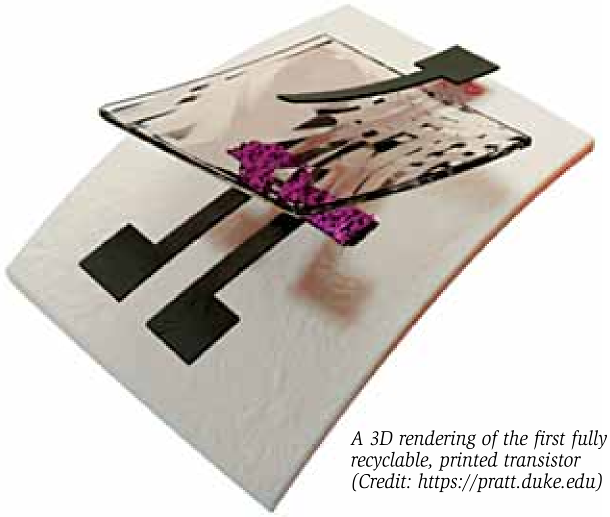
World's first fully recyclable printed electronics

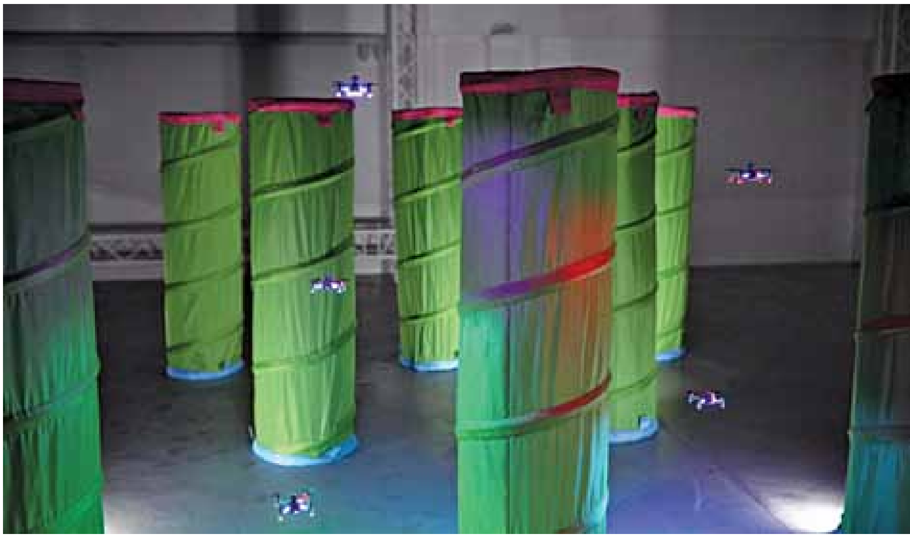


A 3D rendering of the first fully recyclable, printed transistor (Credit: https://pratt.duke.edu)

Engineers at Duke University have developed the world's first fully recyclable printed electronics. By creating a transistor with three carbon based inks, the researchers hope to combat the growing global epidemic of electronics waste. The researchers began by developing a method for suspending crystals of nanocellulose (a biodegradable element) that were extracted from wood fibres that—with the sprinkling of a little table salt—yielded an ink that acts as an insulator in their printed transistors. Using the three inks in an aerosol jet printer at room temperature, the team showed that their all-carbon transistors perform well enough for use in a wide variety of applications, even six months after the initial printing. The technology can be used in a large building needing thousands of simple environmental sensors to monitor its energy use or in customised biosensing patches for tracking medical conditions.

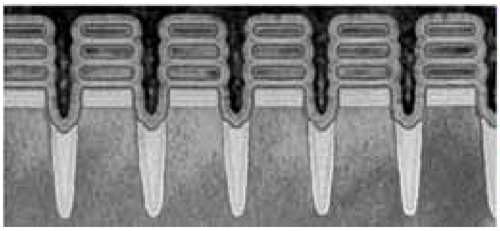
Drones can now fly as swarms quickly and safely

Drone swarms have not been used more widely as it was believed that there is a risk of gridlock or collision within a swarm as each agent has tendency to coordinate its movements with the others, adjusting its trajectory so as to keep a safe inter-agent distance or to travel in alignment. Now engineers at EPFL have developed a predictive control model that allows swarms of drones to fly in cluttered environments quickly and safely. It works by enabling individual drones to predict their own behaviour and that of their neighbours in the swarm. The model enables drones to become less dependent on commands issued by a central computer and instead modify their trajectories autonomously. The model works by programming in locally controlled, simple rules relating to maintenance of minimum inter-agent distance, set velocity and a specific direction to follow.



Engineers at EPFL have developed a predictive control model that allows swarms of drones to fly in cluttered environments quickly and safely. It works by enabling individual drones to predict their own behaviour and that of their neighbours in the swarm (Credit: https://actu.epfl.ch)

Two-nanometre chip technology unveiled by IBM



Two nanometer chips (Credit: https://newsroom.ibm.com)

IBM has unveiled a breakthrough in semiconductor design and process with the development of a 2-nanometre (nm) nanosheet technology. It is projected to achieve 45% higher performance, or 75% lower energy use, than today's most advanced 7nm node chips, thus